Jashore University of Science and Technology, Bangladesh
Department of Computer Science and Engineering
2nd Year 2nd Semester Examination-2024

Course Code: CSE 2201 | Course Title: Data Communication

Time: 3.00 hours | Total marks: 72

Answer any 06 sets of questions from the following 08 sets of questions.

1. a) Define protocols and standards. 2

   b) "A star topology is less expensive than a mesh topology." - Is the statement is true or false? Defend your answer. 3

   c) Assume a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700 and 900 Hz. Calculate its bandwidth and sketch the spectrum, assuming all components have maximum amplitude of 5V. 4

   d) Compare different types of data flows in data communication systems. 3

2. a) Briefly explain different types of addressing used in the TCP/IP protocol suite. 3

   b) A signal travels through a transmission medium and its power is reduced to one-third. Calculate the attenuation of the signal. 2

   c) Assume we need to download text documents at the rate of 100 pages per minute. If each page has 24 lines, 80 characters per line and 8 bits are required to represent a character. Calculate the required bit rate of the channel. 3

   d) Briefly explain signal attenuation and distortion with proper diagrams. 4

3. a) Write the Nyquist bit rate and Shannon capacity theorems in transmissions. 3

   b) Differentiate bandwidth and bit rate of a channel. 2

   c) If the SNR of a 1-MHz bandwidth channel is 63, calculate its bit rate and signal level. 3

   d) Briefly explain the bandwidth-delay product of a channel with a proper example. 4

4. a) Briefly explain the B8ZS scrambling technique with a proper example. 4

   b) A signal is carrying data in which one data element is encoded as one signal element. If the bit rate is 100 kbps and the case factor is 0.5, calculate its baud rate. 2

   c) Compare NRZ-L and NRZ-I line coding methods. 2

   d) Apply Manchester and differential Manchester line coding methods to convert the digital data 010011to digital signal. 4

5. a) What are the differences between parallel and serial transmission? 3

   b) What is the bandwidth of a signal that can be decomposed into five sine waves with frequencies at 0, 20, 50, 100, and 200 Hz? All peak amplitudes are the same. Draw the bandwidth. 2

   c) Define carrier signal and explain its role in analog transmission. 3

d) Which of the four digital-to-analog conversion techniques (ASK, FSK, PSK or QAM) is the most susceptible to noise? Defend your answer. 4

6. a) Define multiplexing and demultiplexing. 2

b) Briefly explain the operation of a FDM. 4

c) Choose the best data rate management strategy in TDM for the following input lines and explain to defend your choice. 6

   i) 20kps, 20kbps, 40kbps, 40kbps, 40kbps
   ii) 50kbps, 25kbps, 25kbps, 25kbps
   iii) 50kbps, 50kbps, 46kbps

7. a) The minimum number of columns in a datagram network is two; the minimum number of columns in a virtual-circuit network is four. Can you explain the reason? Is the difference related to the type of addresses carried in the packets of each network? 4

b) What are the two approaches to packet switching? Briefly explain with necessary diagrams. 3

c) How Multistage Switch differs from Crossbar Switch? Design a three-stage, 200 × 200 switch ($N = 200$) with $k = 4$ and $n = 20$. 5

8. a) Assume the following Hamming codes, C(7,4) are provided in a data communication system. 6

Dataword	Codeword	Dataword	Codeword	Dataword	Codeword	Dataword	Codeword
0000	0000000	0100	0100011	1000	1000110	1100	1100101
0001	0001101	0101	0101110	1001	1001011	1101	1101000
0010	0010111	0110	0110100	1010	1010001	1110	1110010
0011	0011010	0111	0111001	1011	1011100	1111	1111111

Explain the following cases for error detection and correction using the Hamming codes.

   i) The codeword 0100011 is received.
   ii) The codeword 0011001 is received.
   iii) The codeword 0001000 is received.

b) Define the minimum Hamming distance. 2

c) Given the dataword 1010011010 and the divisor 10111. 4

i) Show the generation of the CRC codeword at the sender site (using binary division).

ii) Show the checking of the CRC codeword at the receiver site (assume no error)

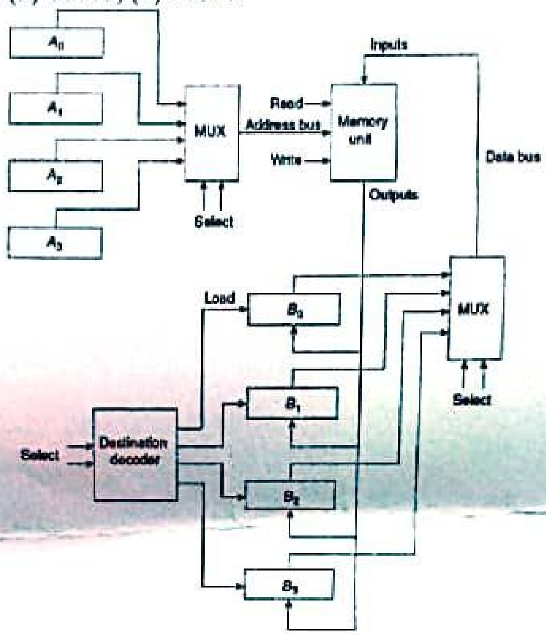
Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2024
2nd Year 2nd Semester, Session: 2021-22

Course Code: CSE 2205 — Time: 3.00 hours
Course Title: Digital System Design — Marks: 72

[N.B. Answer any 06 questions from the following 08 sets of questions. The figures in the right margin indicate full marks.]

1 a) Describe the types of microoperations most often encountered in digital systems. 4

b) Let s1 and s0 be the selection variables for the multiplexer of the following figure, and let d1d0 be the selection variables for the destination decoder. Variable e is used to enable the decoder. State the transfers that occur when the selection variables **s1s0d1d0** are equal to: (1) 00010; (2) O1000; (3) 11100; (4) 01101. 4



c) Illustrate the hardware implementation of the following statement- 4

$$xy'T0 + T1 + x'yT2: A \leftarrow A + B$$

2 a) The following memory transfers are specified for the system of figure 1(b). 6

I. M [A2] ← B3
II. B2 ← M [A3]

Specify the memory operation and determine the binary selection variables for the two multiplexers and the destination decoder.

b) Explain the role of the status register in the Arithmetic-Logic Unit. 4

c) Discuss the usage of the Program Counter in computer design in your own words. 2

3 a) Perform the arithmetic operations (+57) + (-39) and (+57) – (-39) using 6

i) Sign – 2's Complement.
ii) Sign – 10's Complement.

b) Explain the circuit of addition of sign-2's complement numbers with overflow. Differentiate between Logic microoperation and shift microoperation. 6

4 a) What is the difference between these two statements? 4

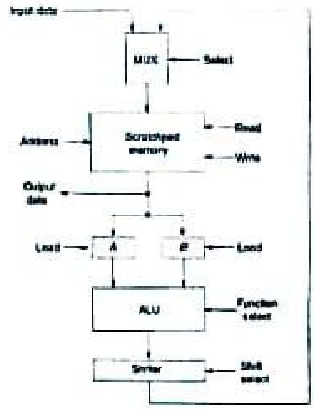
$A + B: F \leftarrow C \vee D$ and $C+D: F \leftarrow A \div B$

b) Design a simple computer to implement the following instructions- 8

Operation code	Mnemonic	Description	Function
00000001	ADD R	Add R to A	$A \leftarrow A + R$
00000010	ADI OPRD	Add operand to A	$A \leftarrow A$ + OPRD
00000011	ADA ADRS	Add direct to A	$A \leftarrow A + M$[ADRS]

5 a) A processor unit employs a scratchpad memory, as the following figure. The processor consists of 64 registers of eight bits each. 4

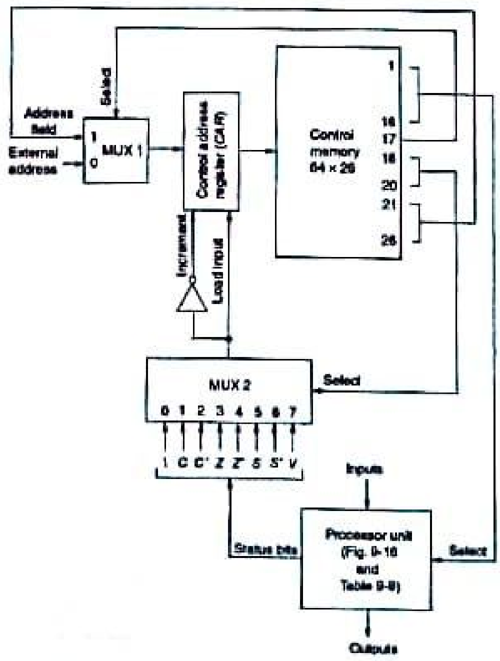
(a) What is the size of the scratchpad memory?
(b) How many lines are needed for the address?
(c) How many lines are there for the input data?
(d) What is the size of the MUX that selects between the input data and the output of the shifter?



b) Design an arithmetic circuit with two selection variables, s1 and s0, that generates the following arithmetic operations. Draw the logic diagram of one typical stage. 8

s_1	s_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A$	$F = A + 1$
0	1	$F = A - B - 1$	$F = A - B$
1	0	$F = B - A - 1$	$F = B - A$
1	1	$F = A + B$	$F = A + B + 1$

6 a) Write a microprogram that compares two unsigned binary numbers stored in R1 and R2. The register containing the smaller number is then cleared. If the two numbers are equal, both registers are cleared. Use the following microprogram system- 4



b) Explain the process of instruction execution in the computer system. 4

i) CMA(Component A) A← A

ii) RLC(Rotate A left with carry)
A←Clc A

iii) RRC (Rotate A right with carry)
A←CRC A

c) Write short notes on 4
1. Hard-wired control.
2. Control of processor unit.
3. Microprogram sequencer.

7 a) Assume the initial value of the accumulator register (A) is $(77B)_{16}$. After executing each of the following instructions, determine the final value of A and update the status bits: 8
 i. i) C (Carry Flag)
 ii. ii) S (Sign Flag)
 iii. iii) Z (Zero Flag)
 iv. iv) V (Overflow Flag)

iv) Exclusive ·NOR to the accumu-lator itself A

b) Explain how the control logic determines the shift direction. 4

8 a) List the register-transfer statements for the execution of the instructions listed below. Assume that the computer does not have an F flip-flop but that the sequence register G has 16 timing variables, t0 through t15. The G register must be cleared when the execution of the instruction is completed. The fetch cycle for the computer now is as follows: 8

t0: PC ← MAR
t1: B ← M, PC ← PC + 1
t2: I ← B(OP)

Each of the following instructions starts the execute cycle from timing variable t3. The last statement must include the microoperation G ← 0.

Symbol	Hexadecimal code	Description	Function
SBA	8 *m*	Subtract from *A*	$A \leftarrow A - M$
ADM	9 *m*	Add to memory	$M \leftarrow A + M$ (*A* does not change)
BEA	A *m*	Branch if *A* equal	If $(A = M)$ then $(PC \leftarrow m)$ (*A* does not change)

b) Differentiate between 4
1. Fetch Cycle & Execute Cycle.
2. Register-reference instructions & Input-output instructions.

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Department of Computer Science and Engineering

Jashore University of Science and Technology

2nd year 2nd Semester, Final Examination 2024

Course Code: MATH-2203 | Time: 3 hours

Course Title: Linear Algebra and Fourier Analysis | Full Marks: 72

[N.B. Answer any 6 set questions from the following 8 set questions. The figures shown in the margin indicate full marks]

1 a) Prove that the matrix $A = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 1+i \\ 1-i & -1 \end{pmatrix}$ is unitary. 3

b) Compute the rank of the matrix, 4

$$\begin{pmatrix} 1 & -1 & 2 & 1 \\ 3 & 0 & 1 & -2 \\ -2 & 1 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

c) Calculate the inverse of the matrix by using row canonical form, 5

$$\begin{pmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{pmatrix}.$$

2 a) Solve the system of linear equations with the help of matrix, 6

$$2x + y - 2z = 10$$
$$3x + 2y + 2z = 1$$
$$5x + 4y + 3z = 4.$$

b) For what value of γ, the following system of linear equations have a solution and solve them completely in each case, 6

$$x + y + z = 1$$
$$x + 2y + 4z = \gamma$$
$$x + 4y + 10z = \gamma^2.$$

3 a) Find the values of λ *and* μ such that the following system has: 6

i. No solutions.

ii. More than one solution.

iii. Unique solution

$$x + y + z = 6$$
$$x + 2y + 3z = 10$$
$$x + 2y + \lambda z = \mu.$$

b) Define eigen value, eigen vector, characteristic polynomial, and characteristic equation of a square matrix. Determine P for 6

$$A = \begin{pmatrix} P & -2 & 1 \\ 2 & -8 & -2 \\ 1 & 2 & 2 \end{pmatrix}.$$

such that $P^{-1}AP$ is a diagonal matrix.

4 a) If V be the set of order pairs (a, b) of real numbers, then examine whether V is a vector space or not over $\mathbb{R}$ with the following operations (addition and scalar multiplication) on V, 5

(i) $(a, b) + (c, d) = (a + d, b + c)$ and $k(a, b) = (ka, kb)$.

(ii) $(a, b) + (c, d) = (ac, bd)$ and $k(a, b) = (ka, kb)$.

b) If $V = \mathbb{R}^3$, $W_1 = \{(a, b, c): a + b + c = 0\}$, and $W_2 = \{(a, b, c): a^2 + b^2 + c^2 \leq 1\}$, then observe whether the sets W_1 and W_2 form a subspace of $\mathbb{R}^3$ or not. 5

c) Define basis and dimension. 2

5 a) What are the idempotent and Nilpotent matrices? Show that $A = \begin{pmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{pmatrix}$ is nilpotent of index P=3. 6

b) Prove that every square matrix can be uniquely expressible the sum of a symmetric and skew symmetric matrix. 6

6 a) Define Fourier series with the physical meaning of the Fourier constants. Write down the Dirichlet's conditions for a Fourier series. 4

b) If an alternating current after passing through a rectifier has the form $i = \begin{cases} I\sin\theta \; for \; 0 < \theta < \pi \\ 0 \quad for \; \pi < \theta < 2\pi \end{cases}$, then find the Fourier series of the function. 4

c) Calculate the Fourier series of the function $f(x)$ defined in $(-2, 2)$ as follows, 4

$$f(x) = \begin{cases} 2 & in -2 \le x \le 0 \\ x & in \; 0 < x < 2 \end{cases},$$

and also sketch this function in $(-6, 6)$.

7 a) Obtain the complex form of the Fourier series of the function, 4

$$f(x) = \begin{cases} 0 & -\pi \le x \le 0 \\ 1 & 0 \le x \le \pi \end{cases}.$$

b) Find the Fourier series as far as the first harmonic to represent the function given by table below, 8

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°
$f(x)$	2.34	3.01	3.69	4.15	3.69	2.20	0.83	0.51	0.88	1.09	1.19	1.64

8 a) What is the integral transform? When an integral transform converts to Fourier complex transform, and Fourier sine transform? State the Fourier integral theorem. 3

b) Calculate the Fourier transform of the functions, 4

$$(i)\; f(x) = \begin{cases} 1 \; for \; |x| < a \\ 0 \; for \; |x| > a \end{cases}$$

$$(ii)\; f(x) = \begin{cases} 1 - x^2 & if \; |x| \le 1 \\ 0 & if \; |x| > 1 \end{cases}.$$

c) State the convolution theorem on Fourier transform. Using Parseval's identity, to prove that $\int_0^\infty \frac{dt}{(a^2+t^2)(b^2+t^2)} = \frac{\pi}{2ab(a+b)}$. 5

Jashore University of Science and Technology, Bangladesh
Department of Computer Science and Engineering
2^{nd} Year 2^{nd} Semester Examination-2024

Course Code: CSE 2203 | Course Title: Database Management System

Time: 3.00 hours | Total marks: 72

Answer any 06 sets of questions from the following 08 sets of questions.

1. a) Define database and database management system. List and explain some applications of DBMS. 4
   b) What are the functions of database languages DDL and DML. How do they differ from each other? 4
   c) Describe different data models that can be used for implementing DBMS. 4

2. a) List the functions of a database administrator. 2
   b) Suppose you are given the following requirements for a simple database for the National Football League (NFL): 10
      - the league has many teams,
      - each team has a name, a city, a coach, a captain, and a set of players,
      - each player belongs to only one team,
      - each player has a name, a position (such as left wing or goalie), a skill level,
      - a team captain is also a player,
      - a game is played between two teams (referred to as host_team and guest_team) and has a date (such as May 11th, 1999) and a score (such as 2 to 4).

   Design a clean and concise ER diagram for the NFL database.

3. a) List different types of constraints used when defining a relation. 2
   b) Consider the following relational database of a college. 10

      Student(RollNumber, StudentName, Address)
      Teachers(TeacherID, TecherName, TeachingSubject)
      College(RollNumber, TeacherID)

   Write Relational algebra expressions for the following requests.
      1. Find the name of Students who live in "Jashore".
      2. Find the name of the teacher who teaches Database Management subject to Rahim.
      3. Insert a new tuple into relation teachers.
      4. Delete records of students whose address is "Pubna".
      5. Rename College relation to University.

4. Consider the following relational schemas, 12

   branch (**branch_name**, brach_city, assets)
   customer (**customer_name**, customer_street, customer_city)
   loan (**loan_number**, branch_name, amount)
   borrower (**customer_name**, **loan_number**)
   account (**account_number**, branch_name, balance)
   depositor (**customer_name**, **account_number**)

   Write SQL expressions for the following queries:

   i. Find the name, loan number, and load amount of all customers having a loan at the Jamalpur branch.
   ii. Find the customer names whose street name starts with the substring "ST%" and having the street name of length 5 characters in total.
   iii. Find all customers name having a loan, an account, or both.
   iv. Find all branches that have greater assets than some branches located in Jamalpur.
   v. Find the loan number of those loans with loan amounts are above and below than $90000 but not equal.
   vi. Find the total loan amount of each branch in Jamalpur city.

5. a) Define functional dependency and trivial functional dependency. 2

b) "If a relation is in BCNF it is in 3NF"- justify the statement. 3

c) Considering the relation R = (A, B, C, G, H, I), compute closure of the functional dependency set, F = { $A \rightarrow B$, $A \rightarrow C$, $CG \rightarrow H$, $CG \rightarrow I$, $B \rightarrow H$} 4

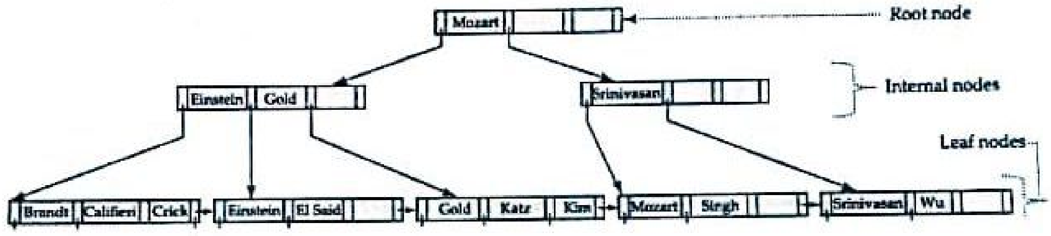
d) Briefly describe the lock-compatibility matrix for concurrency control. 3

6. a) Briefly describe the procedure for storing fixed length records in a database with a proper example. 3

b) Compare dense index and sparse index. 3

c) Consider the following B+ tree and do the following tasks. 6

   i. Insert an index "Adams" to the tree
   ii. Insert an index "Lamport" to the tree
   iii. Delete the index "Srinivasan" from the tree



7. a) Briefly describe the use of a lock table in concurrency control. 3

b) Consider a transaction that transaction transfers 50 tk from account A to account B in a bank. Describe the ACID properties of the transaction. 4

c) List the conditions for view serializability of multiple transactions. 2

d) Find a conflict serializability of the following schedule (if any) and write the conflicting instructions. 3

T_1	T_2
read (A) write (A)	
	read (A) write (A)
read (B) write (B)	
	read (B) write (B)

8. a) Briefly describe the process of generating wait-for graph for deadlock detection in a database management system with a proper example. 3

b) Consider the following logs at three instances of time (t1, t2, t3). Choose the appropriate database recovery phase for each case and explain the recovery process in detail. 6

t1	t2	t3
<T_0 start>	<T_0 start>	<T_0 start>
<T_0, A, 1000, 950>	<T_0, A, 1000, 950>	<T_0, A, 1000, 950>
<T_0, B, 2000, 2050>	<T_0, B, 2000, 2050>	<T_0, B, 2000, 2050>
	<T_0 commit>	<T_0 commit>
	<T_1 start>	<T_1 start>
	<T_1, C, 700, 600>	<T_1, C, 700, 600>
		<T_1 commit>

c) Compare immediate-modification and deferred-modification of databases. 3

P1 - P3 P5 P7
P2 - 3 6 7
P4 - 5 6 7

Department of Computer Science and Engineering

Jashore University of Science and Technology

2nd year 2nd Semester, Final Examination 2024

Course Code: MATH-2201 | Time: 3 hours
Course Title: Probability and Statistics for Engineers | Full Marks: 72

[N.B. Answer any 6 set questions from the following 8 set questions. The figures shown in the margin indicate full marks]

1 a) A tire manufacturer wants to determine the inner diameter of a certain grade of tire. Ideally, the diameter would be 570 mm. The data are as follows: **3+3+2 =8**

572, 572, 573, 568, 569, 575, 565, 570.

i. Find the sample mean and median.
ii. Find the sample variance, standard deviation
iii. Using the calculated statistics in parts (i) and (ii), can you comment on the quality of the tires?

b) The following are historical data on staff salaries (dollars per pupil) for 30 schools sampled in the eastern part of the United States in the early 1970s. **4**

3.79 2.99 2.77 2.91 3.10 1.84 2.52 3.22
2.45 2.14 2.67 2.52 2.71 2.75 3.57 3.85
3.36 2.05 2.89 2.83 3.13 2.44 2.10 3.71
3.14 3.54 2.37 2.68 3.51 3.37

Construct a relative frequency histogram of the data.

2 a) How coefficient of variation (C.V) is used in your practical life? **2**

b) Define standard deviation with example. In what situation standard deviation is suitable instead of coefficient of variation. **2**

c) A purchasing agent obtained samples of 60 watt bulbs from two companies. He had the samples tested in his own laboratory for length of life with the following results: **8**

Length of life (in hours)	Company A	Company B
1800-1900	12	5
1900-2000	18	36
2000-2100	20	21
2100-2200	9	3
2200-2300	6	1

a) Which company's bulbs do you think are better?
b) If prices of both types are the same, which company's bulbs would you buy?

3 a) How many distinct permutations can be made from the letters of the word *INFINITY*? How many of these permutations start with the letter *I*? **2+2=4**

b) If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, 8 or more cars on any given workday are, respectively, 12%, 19%, 28%, 24%, 10% **2x4=8**

i. What is the probability that he will service at least 5 cars on his next day?
ii. What is the probability that no more than 4 cars will be serviced by the mechanic?
iii. What is the probability that he will service fewer than 8 cars?
iv. What is the probability that he will service either 3 or 4 cars?

4 a) Define Skewness and Kurtosis. Explain with examples. **4**

b) For a distribution $\mu_1' = 5$, $\mu_2' = 45$, $\mu_3' = 346$ and $\mu_4' = 18548$. Comment on the shape characteristics of the distribution. **4**

c) The second moment about the mean of three distributions are 4.5, 15.8, and 9.0 while the forth moments about the mean are 43.54, 768.92, and 243.00, respectively. Which of the distribution is — 4

i. Leptokurtic
ii. Mesokurtic
iii. Platykurtic

Radio- 3-1GHz
MScr- 1-300GHz
Infra- 300GHz -400THz

5 a) Define the following terms with examples in the context of probability: — 2x2=4

i. Experiment
ii. Sample Space

b) In the context of a die-tossing experiment, two events are defined as follows: — 4

i. Observing a number greater than 4
ii. Observing an even number

Are these events mutually exclusive? Justify your answer.

c) Which of the following events are equal? — 4

i. $A = \{1, 3\}$;
ii. $B = \{x \mid x \text{ is a number on a die}\}$;
iii. $C = \{x \mid x^2 - 4x + 3 = 0\}$;
iv. $D = \{x \mid x \text{ is the number of heads when six coins are tossed}\}$.

Coaxial - center core
metalic shield
dielectric insulator
plastic jacket
thinnet
thicknet

6 a) Define 'Continuous Random Variable' with an example. — 2

b) Heights of students are normally distributed with mean = 170 cm and standard deviation = 10 cm. Find the probability that a randomly selected student has a height less than 180 cm. — 4

c) Given a standard normal distribution, find the value of k such that — 3+3=6

Twisted pair - UTP
STP

i. $P(Z>k)=0.3015$
ii. $P(k<Z<-0.18) = 0.4197$

Fiber optic - core
cladding
outer jacket

7 a) Define Bivariate Data. — 2

b) What kind of relationship Bivariate Data may show? Explain with examples. — 4

c) The grades of a class of students midterm report (x) and on the final examination (y) areas follows: — 3+3=6

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

i. Estimate The Linear Regression Line.
ii. Estimate The Final examination final examination grade of a student who received 85 on the midterm report.

8 a) Define poison distribution. Give some real example where poison distribution uses. — 2

b) If the probability of a car accident in a very busy road in an hour is 0.001. If 2000 cars passed in one hour by that road, what is the probability that — 4

i. Exactly 3
ii. More than two car accidents happened on that hour of the road.

c) The probability that a MBA graduate from a public university gets an executive job is 0.6. Four MBA graduates applied for executive jobs. What is the probability that — 6

i. Exactly 3 will get the job?
ii. At least 2 will get the job?
iii. At least 3 will get the job?
iv. None will get the job?

Department of Computer Science & Engineering

Jashore University of Science and Technology

2^{nd} Year 2^{nd} Semester, Final Examination, 2024

Course Code: HUM 2201 | Course Title: Business Psychology

Full mark: 72 | Time: 3 hours

[N.B. Answer any 6 set questions from the following 8 set questions. The figures shown in the margin indicate full marks]

1 a) Define the key concepts of business psychology. How does a student from CSE (Computer Science and Engineering) benefit from business psychology course? 6

b) Differentiate between hard and soft skills. Today's organizations are shaped by various challenges and opportunities. Describe the following factors that have a general impact on organizations: social media, technology, workforce diversity, and ongoing globalization. 6

2 a) Differentiate between surface level and deep level diversity. What are the benefits of having age diversity in an organization? 6

b) Define culture. Apply the famous Hofstede's cultural framework to evaluate the culture of our country (Bangladesh). 6

3 a) Define attitude. What are the components of attitude. 5

b) Define job satisfaction. "Do you think that job satisfaction has some specific outcomes"- if yes what are those outcomes? 7

4 a) Define emotional intelligence. Draw the model of emotional intelligence and explain it. 6

b) Analyze how emotions and moods affect selections process, creativity, leadership, safety, and injury at work. 6

5 a) Define person job fit and person organization fit. Explain the Holland's topology of personality and congruent occupations. 6

b) The Myerrs-Briggs Type Indicator (MBTI) is the most widely used personality assessment instrument in the world. It is a one hundred question personality test that asks people how they usually feel or act in situations. Discuss the four four dimensions of MBTI personality model. 6

6 a) Explain the Big Five personality model with suitable example. 4

b) The three negative personality qualities known as the "Dark Triad" are psychopathy, Machiavellianism, and narcissism. Give appropriate examples to illustrate the trio. 4

c) How do the concepts of core self-evaluation (CSE), self-monitoring, and proactive personality help us understand personality? 4

7 a) Briefly explain the Maslow's hierarchy of needs theory. 4

b) How to use McClelland's theory of needs to motivate employees? Explain. 4

c) "Expectancy theory argues that the strength of our tendency to act a certain way depends on the strength of our expectation of a given outcome and its attractiveness."- In line with the statement, explain the three factors of expectancy theory. 4

8 a) Differentiate between a manger and a leader. 4

b) Define charismatic leadership theory. Does effective charismatic leadership depend on situation? 8